Advanced SQL

Exercise 2

DATE Function

Q1 ▸

```
SELECT  employee_id,
        name,
        hire_date
DATEADD (month, 6, hire_date) AS hire_plus_6_months
FROM    Employees;
```

Outcome.

employee_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Q2 →

```
SELECT Student_id,
       name
DATEDIFF (day, dob, Current_date) AS age_in_days
FROM Students;
```

Outcome

Student_id	name	age_in_days
101	Maya	7560
102	Ethen	7782
103	Sienna	7343

Q3 →

```
SELECT event_id,
       event_name
DATEDIFF(day, Current_date, event_date) AS days_remaining
FROM Events;
```

Outcome

event_id	event_name	days_remaining
1	Seminar	-676
2	workshop	-234
3	Hackathon	-457

Q4 →

```
SELECT invoice_id,
       issue_date,
       due_date
DATEDIFF (day, issue_date, due_date) AS days_between
FROM invoices;
```

Outcome

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Q5

```
SELECT Course_id,
       name
TO_CHAR (start_date, 'Month YYYY') AS formatted_date
FROM Courses;
```

Outcome

Course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

Q6

```
SELECT Member_Id
DATE_FROM_PARTS (Start_year, Start_month, Start_day) AS full_Start_date
FROM Memberships;
```

Output

Member_Id	full_Start_date
1	2023-5-16
2	2022-11-25

Q7.

```
SELECT Sub_Id,
       Plan
DATEADD (Year, 1, renewal_date) AS extented_renewal_date
FROM Subscriptions;
```

Output

Sub_Id	plan	extended_renewal_date
11	Basic	2026-0101
12	Premium	2026-03-15

Q8

```
SELECT order_id,
       order_date,
       Current_date AS today_date
DATEDIFF (day, order_date, Current_date) AS day_since_order
FROM Orders;
```

Output

Order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-22	372
1002	2025-04-10	2026-04-22	377

Q9.

```
SELECT training_id,
       topic
DATE_PART (Year, training_date) AS training_Year
FROM Trainings;
```

Output

training_id	topic	training_Year
1	Safety	2025
2	Complience	2025

Q10.

```
SELECT  post_id,
        title
EXTRACT (hour FROM published_on) AS hour_published,
EXTRACT (Minute FROM published_on) AS minute_published,
FROM Blog_posts;
```

Output

Post_id	title	hour_published	minute_published
1	SQL tips	10	15
2	Data cleaning	16	45

Q11

```
SELECT driver_id,
       license_expiry
DATEDIFF (day, current_date, license_expiry) AS days_left
FROM Drivers;
```

Output

driver_id	license_expiry	days_left
301	2025-08-10	-255
302	2023-12-31	-843

Q12.

```
SELECT    message_id,
          Sent_timestamp
DATEDIFF (second, Sent_timestamp, Current_timestamp) As Second_Since_Sent;
FROM Messages;
```

Output

Message_id	Sent_timestamp	Current_timestamp	Seconds_Since_Sent
1	2025-04-19 09:32:45	2026-04-22 HH:MM:SS	-33,000,000
2	2025-04-18 23:59:59	2026-04-22 HH:MM:SS	-33,050,000

Q13.

```
SELECT  return_id,
        return_date
DATEADD (day, 15, return_date)   restock_date
FROM Returns;
```

Output

return_id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-06

Q14

```
SELECT assign_id
TO_DATE (assigned_on, 'YYYY-MM-DD') AS assigned_on_date
FROM Assignment;
```

Output

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

Q15

```
SELECT meeting_id
TO_CHAR(sheduled_time, 'FMMonth DD, YYYY "at" HH:MM:SS') AS format
FROM Meetings;
```

Meeting_id	formatted_meeting_time
1	April 19, 2025 at 2:00 PM
2	April 19, 2025 at 9:30 AM